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Supplementary Information

A OMat24 psuedopotential differences

Differences between OMat24 DFT and Materials Project PBE calculations are largely attributed to different versions of pseudo-potentials and different pseudo-potential symbols for Yb and W. In OMat24 we use version 54 of VASP pseudopotentials, while the Materials Project calculations use an older releases of VASP pseudopotentials⁵¹. Table A.1 lists element symbols and pseudopotential generation dates for all elements with different pseudo potentials.

Element	MP (version 52)	OMat (version 54)
Cl	Cl 17Jan2003	Cl 06Sep2000
S	S 17Jan2003	S 06Sep2000
Au	Au 06Sep2000	Au 04Oct2007
W	W_pv 06Sep2000	W_sv 04Sep2015
Rh	Rh_pv 06Sep2000	Rh_pv 25Jan2005
Pt	Pt 05Jan2001	Pt 04Feb2005
Ru	Ru_pv 06Sep2000	Ru_pv 28Jan2005
Pd	Pd 05Jan2001	Pd 04Jan2005
P	P 17Jan2003	P 06Sep2000
As	As 06Sep2000	As 22Sep2009
Mo	Mo_pv 08Apr2002	Mo_pv 04Feb2005
Ge	Ge_d 06Sep2000	Ge_d 03Jul2007
Ag	Ag 06Sep2000	Ag 02Apr2005
Tc	Tc_pv 06Sep2000	Tc_pv 04Feb2005
Co	Co 06Sep2000	Co 02Aug2007
Fe	Fe_pv 06Sep2000	Fe_pv 02Aug2007
Ga	Ga_d 06Sep2000	Ga_d 06Jul2010
Cr	Cr_pv 07Sep2000	Cr_pv 02Aug2007
Mn	Mn_pv 07Sep2000	Mn_pv 02Aug2007
Zr	Zr_sv 07Sep2000	Zr_sv 04Jan2005
Mg	Mg_pv 06Sep2000	Mg_pv 13Apr2007
Y	Y_sv 06Sep2000	Y_sv 25May2007
Gd	Gd 08Apr2002	Gd 23Dec2003
Eu	Eu 08Apr2002	Eu 23Dec2003
Ce	Ce 28Sep2000	Ce 23Dec2003
Yb	Yb_2 06Sep2000	Yb_3 08Jul2013
Li	Li_sv 23Jan2001	Li_sv 10Sep2004
Na	Na_pv 05Jan2001	Na_pv 19Sep2006

Table A.1 OMat24 and MP POTCAR file differences. POTCAR file symbols and titel dates for all elements that with different POTCARs in Materials Project and OMat24 PBE calculations.

B Dataset statistics

Figure B.1 shows histograms for the number of atoms per structure in the sub-datasets that make up the OMat24 dataset. Similarly Figure B.2 shows the distributions of energy, forces and stress in the different sub-datasets. The number of atoms per system is shifted to largers sizes, since these structures were generated by tiling the unit cell of relaxed Alexandria structures. The rest of the distributions are concentrated at smaller sizes, as a direct result of the distribution in the Alexandria dataset shown in Figure 1b in the main text.

From Figure B.2 we can see that the AIMD portion of the dataset has a much narrower force and stress